S^{2}Mamba: A Spatial–Spectral State Space Model for Hyperspectral Image Classification

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***Abstract*— The land cover analysis using hyperspectral images (HSIs) remains an open problem due to their low spatial resolution and complex spectral information. Recent studies are primarily dedicated to designing Transformer-based architectures for spatial–spectral long-range dependencies modeling, which is computationally expensive with quadratic complexity. Selective structured state space model (SSM; Mamba), which is efficient for modeling long-range dependencies with linear complexity, has recently shown promising progress. However, its potential in HSI processing that requires handling numerous spectral bands has not yet been explored. In this article, we innovatively propose S^{2}Mamba, a spatial–spectral SSM for HSI classification, to excavate spatial–spectral contextual features, resulting in more efficient and accurate land cover analysis. In S^{2}Mamba, two selective structured SSMs through different dimensions are designed for feature extraction, one for spatial, and the other for spectral, along with a spatial–spectral mixture gate (SMG) for optimal fusion. More specifically, S^{2}Mamba first captures spatial contextual relations by interacting each pixel with its adjacent through a patch cross scanning (PCS) module and then explores semantic information from continuous spectral bands through a bidirectional spectral scanning (BSS) module. Considering the distinct expertise of the two attributes in homogenous and complicated texture scenes, we realize the SMG by a group of learnable matrices, allowing for the adaptive incorporation of representations learned across different dimensions. Extensive experiments conducted on HSI classification benchmarks demonstrate the superiority and prospect of S^{2}Mamba. The code will be made available at: https://github.com/PURE-melo/S2Mamba.**

***Index Terms*— Hyperspectral image (HSI) classification, linear complexity, spatial–spectral feature fusion, spatial–spectral state space model (SSM).**

I. Introduction

HYPERSPECTRAL images (HSIs), consisting of numerous spectral bands, are capable of land cover analysis due to their rich material information [1], [2], [3], [4], [5], with extensive application in domains such as precision agriculture, mineral exploration, and environmental monitoring [6], [7]. Therefore, there is a strong incentive to design a more effective and efficient model for HSI classification. Convolutional neural networks (CNNs) [8], [9], [10], [11], [12] as a widespread paradigm have been widely studied in HSI classification. However, this paradigm is limited by its local receptive fields and is unable to comprehensively capture continuous spectral properties. Transformer architectures [13], [14], [15], [16], [17], [18] have been recently explored in HSI classification, which exhibits remarkable performance due to their ability to extract global contextual information on both spatial and spectral dimensions. Thanks to their powerful capability, several transformer-based spectral foundation models [4], [5] have been presented.

Despite its powerful representation ability, dealing HSI data with Transformer-based models is computationally expensive, primarily due to the self-attention mechanism with quadratic computational complexity $\mathcal{O}(N^2)$ [19], [20]. As an efficient alternative to self-attention mechanisms, the selective structured state space model (SSM; Mamba) [21] has recently emerged as a powerful tool with linear complexity for modeling long-range dependency in sequence processing. Derived from this, a series of Mamba-based models [22], [23], [24], [25], [26], [27], [28], [29], [30] have been explored for various computer vision tasks, such as image classification, semantic segmentation, and others. However, most of these models are only applied in natural image processing, leaving a blank in HSI classification due to the difficulty in handling complicated spatial–spectral information.

Drawing inspiration from the success of SSMs, this article seeks to explore their potential for HSI processing. To this end, we propose a spatial–spectral SSM, termed S^{2}Mamba, to jointly excavate long-range spatial relations and continuous spectral features for HSI classification. Our S^{2}Mamba comprises patch cross scanning (PCS) and bidirectional spectral scanning (BSS) modules for capturing spatial and spectral information, respectively, and merging them with a spatial–spectral mixture gate (SMG). Specifically, we build a PCS mechanism to capture the contextual relations between adjacent pixels, wherein the patch data are first flattened into pixel sequences through different route generation ways and a selective structured SSM is then applied to these sequences for capturing contextual features. Considering the rich knowledge within continuous spectral bands, we design an extra scanning module on the spectral dimension for retrieving semantic properties in HSI data by a bidirectional interaction between each band.

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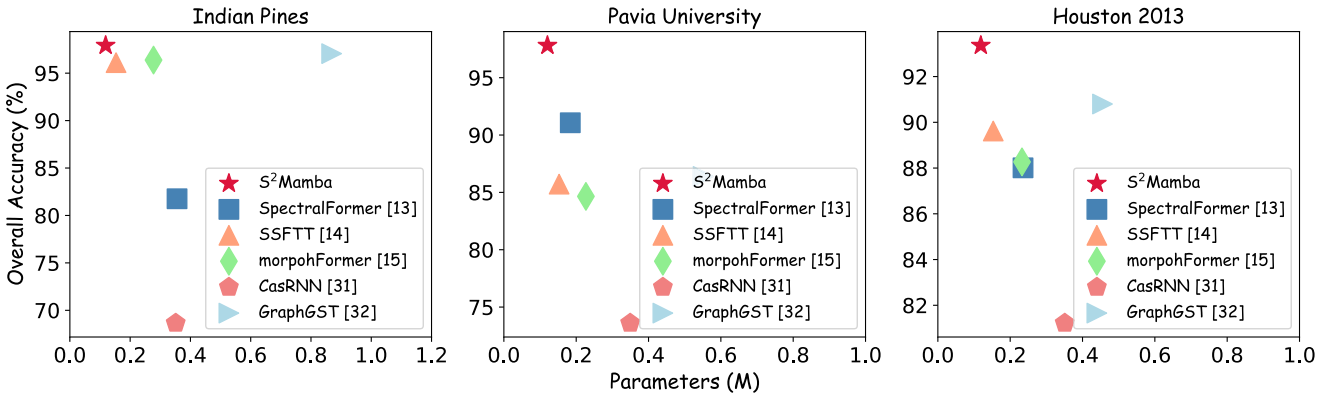

Fig. 1. OA (%) and parameters (M) comparison on IP, PU, and Houston 2013 datasets. Our proposed S^{2}Mamba achieves optimal results in terms of OA and parameters compared with existing methods [13], [14], [15], [31], [32].

The remaining dilemma lies in optimally merging spatial and spectral attributes of HSI data. We observe that spectral information exhibits a more significant role for uniform land cover regions than those regions with complex textures. This is due to the paucity of spatial cues within homogenous regions, necessitating the reliance on spectral information for determining land cover categories. In contrast, the spatial information can offer a considerable prior in dealing with regions containing complex textures, thereby enhancing the classification performance. To address this, we here propose an SMG to dynamically merge the above features pixel-by-pixel, wherein each spatial location is assigned a group of learnable weights to determine the ratio of various features. This gating mechanism, promoting the competition of the two features, facilitates the integration of spatial–spectral attributes by truncating those redundant ones, greatly boosting the accuracy.

Through experimental assessment, we verify the effectiveness of S^{2}Mamba on the three public HSI classification datasets. Fig. 1 illustrates that our S^{2}Mamba improves the previous SOTA method by 0.86%, 6.74%, and 2.56% in terms of overall accuracy (OA) on Indian Pines (IP), Pavia University (PU), and Houston 2013 datasets, respectively. Meanwhile, it outperforms Transformer-based models with the fewest parameters (about 0.12 M) and a linear complexity $\mathcal{O}(kN)$, where $k \ll N$.

The contribution of our work can be summarized as follows.

1) A novel spatial–spectral SSM, termed S^{2}Mamba, is proposed for HSI classification, which is the innovative exploration to introduce Mamba into hyperspectral classification. Extensive experiments demonstrate that S^{2}Mamba can achieve state of the art with the fewest parameters.
2) A PCS module is designed to excavate spatial contextual relations between adjacent pixels by flattening HSI data into multiple pixel sequences and scanning them with a selective structured SSM, thereby facilitating land cover discrimination.
3) Considering the rich prior within spectral, a BSS module is proposed to fully explore the spectral discriminative properties by scanning continuous bands with a selective structured SSM, enhancing the classification capability of the model.
4) To optimally integrate spatial and spectral attributes of HSI, an SMG is proposed, which controls the domination of the two features by adaptive feature competition, further ameliorating the model performance.

The remainder of this article is organized as follows. In Section II, we briefly review related works on HSI classification and SSMs. In Section III, we describe the details of the proposed HSI classification framework S^{2}Mamba. We introduce the dataset and evaluation metrics in Section IV, and show the experimental settings and results in Section V. Finally, Section VI concludes this article.

II. Related Works

A. HSI Classification

Existing methods typically solve the HSI classification problem by leveraging well-crafted deep neural networks, which can be primarily divided into several categories: CNN-based [8], [9], [10], [11], [12], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], recurrent neural network (RNN)-based [31], [48], [49], [50], and Transformer-based [13], [14], [15], [16], [17], [18], [32], [51], [52], [53], [54].

1) Traditional Models: CNNs are widely applied in hyperspectral classification tasks, an excellent feature extraction architecture that captures spatial and local semantic information. Recent works have explored CNNs to individually extract spatial [8], [37] and spectral features [38], [39], [47] from hyperspectral remote sensing images, or to learn spatial–spectral joint representation [9], [10], [33], [34], [35], [36], [45], achieving remarkable progress. Considering that CNN structures are incapable of modeling irregular data, some research introduced graph convolutional networks (GCNs) [11], [12], [40], [41], [44], [46] to mine the potential spatial semantic information of HSI data. Another group of studies serves the spectral information across different bands of HSI data as continuous sequences and employs RNNs [31], [48], [49], [50] to extract spectral features for classification. However, these methods often struggle to extract global spectral information due to their limited capability in long-range dependency.

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2) Transformer-Based Models: Transformer is a powerful architecture consisting of multiple self-attention mechanisms to extract global contextual information, and currently, they have been explored for HSI classification. Most of these methods [13], [14], [15], [16], [17], [18], [32], [51], [52], [53], [54] attempt to learn the global sequential information on both spatial and spectral dimensions. Spectralformer [13] is the first one in its kind to introduce the Transformer architecture into HSI classification, which jointly captures the local and global information by grouping the adjacent bands. Beyond pure Transformer structures, some methods adopt hybrid networks to acquire spatial–spectral features. SSFTT [14] utilizes convolutional layers to describe the low-level features and integrate them through Transformer layers. morphFormer [15] uses morphological convolution layers to learn both spatial and spectral representations and merges them by applying Transformer layers. In addition, other works [54] incorporate RNNs with Transformers to collaboratively extract continuous spectral features and spatial context features, enhancing the classification performance.

Unlike previous methods that capture long-range dependencies by compute-intensive Transformer structures, we innovatively explore a fully sequential architecture based on selective structured SSMs for efficient global spatial–spectral feature extraction.

B. State Space Models

Most recently, SSMs [55], [56], [57], [58], [59], especially the structured state space sequence models (S4) [56] have shown promising progress in sequence analysis, which are capable of long-range sequence modeling with linear computational complexity. By introducing a selective mechanism into [56], Mamba [21] further optimizes its context compression ability and exhibits superior performance to Transformers. Considering its outstanding performance in sequence data processing, many works have explored the potential of Mamba [21] in computer vision and achieved promising advancements. In particular, visual SSM (Vmamba) [23] and vision Mamba (Vim) [22] have recently emerged as powerful tools for various computer vision tasks due to their efficiency in modeling long-range dependencies. On the basis of them, a series of visual SSMs have been proposed, such as medical image analysis [26], [28], [60], [61], video understanding [29], [62], and others [24], [25], [27], [30], [63], [64], [65]. However, most of these methods are only applied in RGB images, leaving a blank in HSI classification that requires handling complex spectral information. Thus, we propose a Mamba-based architecture for HSI classification, fully exploiting spatial–spectral features by SSMs.

III. Proposed Method

A. Preliminaries

1) State Space Models: SSMs are foundational statistical models that describe the dynamic behavior of systems and are widely applied in domains such as time-series analysis and control systems. In SSMs, the continuous evolution of systems is calculated through a set of ordinary differential equations (ODEs), which map an input signal into a latent space and decode it into an output sequence. This operation can be defined as follows:

$$\begin{aligned} h'(t) &= \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) &= \mathbf{C}h(t) + \mathbf{D}x(t) \end{aligned} \tag{1}$$

where $h(t) \in \mathbb{R}^N$, $x(t) \in \mathbb{R}^L$, and $y(t) \in \mathbb{R}^L$ indicate the latent state, input signal, and output signal, respectively. $h'(t) \in \mathbb{R}^N$ refers to the time derivative of $h(t)$. N and L represent the dimensions of the latent space and sequences, respectively. In addition, $\mathbf{A} \in \mathbb{R}^{N\times N}$ is the state transition matrix. $\mathbf{B} \in \mathbb{R}^{N\times L}$ and $\mathbf{C} \in \mathbb{R}^{L\times N}$ are projection matrices. $\mathbf{D}$ is usually served as a residual connected operation and thus discarded in equations. To facilitate continuous SSMs with discrete sequences, Mamba [21] employs a zero-order hold technique to discretize the ODEs

$$\begin{aligned} \overline{\mathbf{A}} &= \exp(\Delta\mathbf{A}) \\ \overline{\mathbf{B}} &= (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I}) \cdot \Delta\mathbf{B} \end{aligned} \tag{2}$$

where $\overline{\mathbf{A}}$ and $\overline{\mathbf{B}}$ represent the discretized forms of the parameters $\mathbf{A}$ and $\mathbf{B}$, respectively, through a discretization step size Δ. As stated in [21], the projection matrix $\overline{\mathbf{B}}$ can be approximated using the first-order Taylor series

$$\overline{\mathbf{B}} = (\exp(\mathbf{A}) - I)\mathbf{A}^{-1}\mathbf{B} \approx (\Delta\mathbf{A})(\Delta\mathbf{A})^{-1}\Delta\mathbf{B} = \Delta\mathbf{B}. \tag{3}$$

After discretization, the ODEs of SSMs can be represented as follows:

$$\begin{aligned} h_t &= \overline{\mathbf{A}}h_{t-1} + \overline{\mathbf{B}}x_t \\ y_t &= \mathbf{C}h_t. \end{aligned} \tag{4}$$

2) Selective Scan Mechanism: Traditional SSMs are linear time-invariant, i.e., the projection matrices do not vary with input signals, resulting in nonselective attention on each sequence unit. To alleviate this, Mamba [21] modifies the parameter matrices to be input-dependent, i.e., $\mathbf{B} \in \mathbb{R}^{B\times L\times N}$, $\mathbf{C} \in \mathbb{R}^{B\times L\times N}$, and $\Delta \in \mathbb{R}^{B\times L\times D}$ are calculated based on $\mathbf{X} \in \mathbb{R}^{B\times L\times D}$, further improving the ability for handling complex sequences by transforming the SSMs into linear time-varying systems.

B. S²Mamba

1) Overview: Fig. 2(a) illustrates the overall network architecture of our proposed S²Mamba framework for HSI classification. Given the input data $\mathbf{X} \in \mathbb{R}^{P\times P\times K}$, where P and K denote the patch size and spectral band number of the data cube, respectively, which is first fed into a 1×1 convolutional layer for embedding. After this process, we can obtain the encoded feature $\mathbf{E} \in \mathbb{R}^{P\times P\times D}$, where D denotes the latent dimension. Then, we feed $\mathbf{E} \in \mathbb{R}^{P\times P\times D}$ into PCS and BSS modules to capture spatial and spectral features, respectively. Next, the above terms are fused by an SMG for fusing. The structure of the SMG is detailed in Fig. 2(d), which simply comprises two fully connected layers with activation functions. Finally, the fused spatial–spectral features are fed into a fully connected layer for category prediction.

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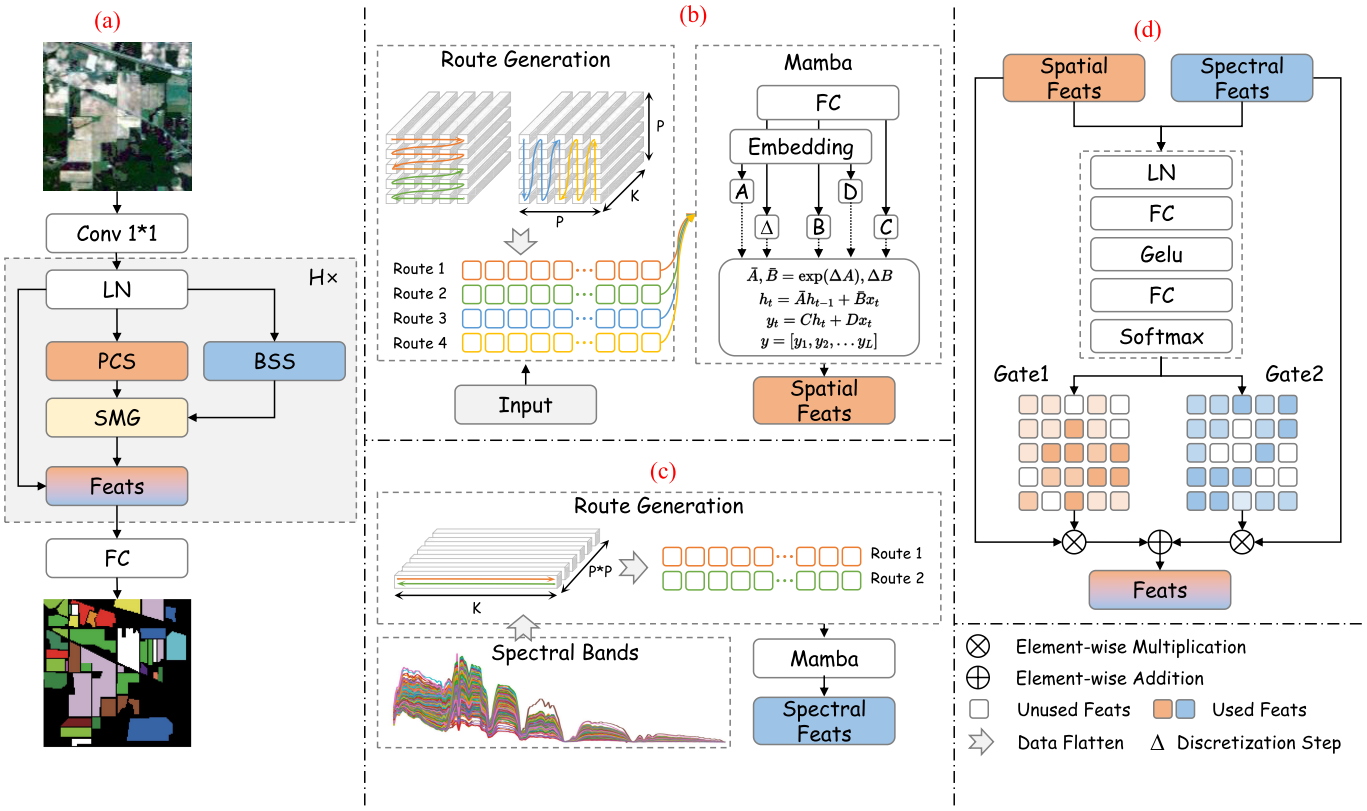

Fig. 2. Illustration of our proposed S^{2}Mamba, consisting of a PCS mechanism, a BSS mechanism, and an SMG. (a) Main framework. (b) PCS. (c) BSS. (d) SMG.

Notably, our S^{2}Mamba is extremely lightweight but efficient, where the number of layers H is only set to 1, and the latent dimension D is set to 64. Moreover, unlike recent advanced approaches that utilize Transformer-based networks with quadratic complexity to capture global spatial–spectral features, our S^{2}Mamba is a powerful spatial–spectral information extraction network with the linear computational complexity.

2) PCS Mechanism: To facilitate the selective scanning mechanism with HSI inputs, we first extend the vanilla selective scan mechanism [21], [23] to the patch-level HSI data and design a PCS mechanism, which captures spatial contextual relations by interacting each pixel with its adjacent through an SSM. Given the input of HSI patch $\mathbf{X} \in \mathbb{R}^{P\times P\times K}$, where P and K denote the patch size and spectral band number of the data cube, respectively, we perform a pixel-by-pixel scanning strategy on four different routes. As illustrated in Fig. 2, each route is generated from different directions, such as top to bottom, left to right, and vice versa.

More specifically, we first flatten the patch data into 1-D sequences with the preset routes, and then recurrently calculate each item of sequences $\mathcal{X}_{\text{seq}} = \{[\mathbf{x}_0^i, \mathbf{x}_1^i, \ldots, \mathbf{x}_{P\times P}^i] \ | \mathbf{x}_j^i \in \mathbb{R}^{1\times K}, i \in \{0, 1, 2, 3\}\}$ by using rewritten (4)

$$\begin{aligned} \mathbf{h}_j^i &= \overline{\mathbf{A}}_{\text{spa}}\mathbf{h}_{j-1}^i + \overline{\mathbf{B}}_{\text{spa}}\mathbf{x}_j^i \\ \mathbf{y}_j^i &= \mathbf{C}_{\text{spa}}\mathbf{h}_j^i + \mathbf{x}_j^i \end{aligned} \tag{5}$$

where $\overline{\mathbf{A}}_{\text{spa}}$, $\overline{\mathbf{B}}_{\text{spa}}$, and $\mathbf{C}_{\text{spa}}$ represent the trainable parameters in PCS. After scanning, we can obtain a set of output sequences

$$\mathcal{Y}_{\text{seq}} = \left\{ \left[\mathbf{y}_0^i, \mathbf{y}_1^i, \ldots, \mathbf{y}_{P\times P}^i\right] \ | \mathbf{y}_j^i \in \mathbb{R}^{1\times K}, i \in \{0, 1, 2, 3\} \right\}. \tag{6}$$

Next, the output sequences obtained from different scanning routes are fused according to the operations in Fig. 2, such as flipping or transposing sequences. As a result, each element in the output sequence $\mathbf{Y} \in \mathbb{R}^{P\times P\times K}$ can integrate influences with its adjacent regions from different directions.

3) BSS Mechanism: Although the above scanning mechanism involves the spatial contextual information within the data cube, it lacks consideration for continuous spectral band information inherent in HSI data. One naive approach to address this issue is to scan the data cube band by band, capturing the semantic cues from rich spectral bands. However, due to the unidirectional information induction attribute of SSMs, the spectral scanning mechanism conducted in a single direction may fail to adequately capture the contextual information between spectral bands, leading to limited spectral utilization.

To this end, we further design a BSS mechanism, analyzing the varying tendency of the continuous spectrum from multiple directions by scanning spectral dimension band by band. We first flatten the HSI patch along the spatial dimensions to acquire the data matrix $\mathbf{S} \in \mathbb{R}^{K\times P^2}$, and then recurrently calculate each item of sequences $\mathcal{S}_{\text{seq}} = \{[\mathbf{s}_0^i, \mathbf{s}_1^i, \ldots, \mathbf{s}_K^i] \ | \mathbf{s}_j^i \in \mathbb{R}^{1\times P^2}, i \in \{0, 1\}\}$ by the following operation:

$$\begin{aligned} \tilde{\mathbf{h}}_j^i &= \overline{\mathbf{A}}_{\text{spe}}\tilde{\mathbf{h}}_{j-1}^i + \overline{\mathbf{B}}_{\text{spe}}\mathbf{s}_j^i \\ \mathbf{p}_j^i &= \mathbf{C}_{\text{spe}}\tilde{\mathbf{h}}_j^i + \mathbf{s}_j^i \end{aligned} \tag{7}$$

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where $\mathbf{p}_j^i$ denotes the jth element in the ith order of output sequences. $\overline{\mathbf{A}}_{\text{spe}}$, $\overline{\mathbf{B}}_{\text{spe}}$, and $\mathbf{C}_{\text{spe}}$ represent the trainable parameters in BSS. After scanning, we can obtain a set of output sequences

$$\mathcal{P}_{\text{seq}} = \left\{ \left[\mathbf{p}_0^i, \mathbf{p}_1^i, \ldots, \mathbf{p}_K^i\right] \mid \mathbf{p}_j^i \in \mathbb{R}^{1\times P^2}, i \in \{0,1\} \right\}. \quad (8)$$

Next, the output sequences $\mathbf{P} \in \mathbb{R}^{P\times P\times K}$ are fused from different routes, which integrate influences from adjacent bands into each spectral band, further boosting its discriminative ability.

4) Spatial–Spectral Mixture Gate: After acquiring the spatial and spectral information of HSI through the two scanning modules, calculating the optimal mixture representation becomes a key challenge. As discussed in Section I, the effectiveness of spatial and spectral features in HSI classification differs across different scenarios, thereby their direct merging without prior may lead to contradictions.

Specifically, we note that spectral information acts more prominence in the classification of uniform regions as opposed to those characterized by complicated textures. This is attributable to the scarcity of spatial cues within homogenous regions, which in turn emphasizes the importance of spectral information for differentiating these land cover. In this case, PCS may contribute to misleading directions as a consequence of the redundant features. Inversely, it offers considerable prior in dealing with regions characterized by complex textures, thereby enhancing the richness of discriminative representation.

To address this, we here propose an SMG to dynamically merge the above features for each position, wherein each location is assigned a group of learnable weights $\tilde{\mathbf{M}} \in \mathbb{R}^{P\times P\times 2}$ to determine the ratio of various features as follows:

$$\begin{aligned}\tilde{\mathbf{M}}_0 &= \frac{\exp(\mathcal{H}(\mathbf{Y};\Theta_g))}{\exp(\mathcal{H}(\mathbf{Y};\Theta_g)) + \exp(\mathcal{H}(\mathbf{P};\Theta_g))} \\ \tilde{\mathbf{M}}_1 &= \frac{\exp(\mathcal{H}(\mathbf{P};\Theta_g))}{\exp(\mathcal{H}(\mathbf{Y};\Theta_g)) + \exp(\mathcal{H}(\mathbf{P};\Theta_g))}\end{aligned} \quad (9)$$

where $\mathcal{H}(*;\Theta_g)$ indicates the feature encoder in SMG. It consists of two fully connected layers with a Gaussian error linear unit activation function. Next, a softmax activation function is applied to transfer them into probability maps, whose values are between 0 and 1. Subsequently, we merge the above features as follows:

$$\mathbf{F} = (\tilde{\mathbf{M}}_0 \cdot \mathbb{1}(\tilde{\mathbf{M}}_0 > \tau)) \odot \mathbf{Y} + (\tilde{\mathbf{M}}_1 \cdot \mathbb{1}(\tilde{\mathbf{M}}_1 > \tau)) \odot \mathbf{P} \quad (10)$$

where τ is the threshold for pruning those low contribution features. By employing this gating mechanism, redundant features that do not contribute to HSI classification can be effectively truncated, facilitating the integration of spatial–spectral attributes. The intuition behind SMG is to encourage competition between the two types of features to select the most discriminative ones under different scenarios, thereby achieving a more satisfactory fusion.

5) Computational Complexity Analysis: Given the HSI patch data with K spectral bands and N pixels, the proposed S²Mamba encloses four spatial scanning routes for pixels and two spectral scanning routes for spectral bands, respectively.

TABLE I
LAND-COVER CATEGORIES OF THE IP DATASET AND ITS CORRESPONDING TRAINING AND TESTING SAMPLES

No.	Category	Training	Testing	Color
1	Corn Notill	50	1384	
2	Corn Mintill	50	784	
3	Corn	50	184	
4	Grass Pasture	50	447	
5	Grass Trees	50	697	
6	Hay Windrowed	50	439	
7	Soybean Notill	50	918	
8	Soybean Mintill	50	2418	
9	Soybean Clean	50	564	
10	Wheat	50	162	
11	Woods	50	1244	
12	Buildings Grass Trees Drives	50	330	
13	Stone Steel Towers	50	45	
14	Alfalfa	15	39	
15	Grass Pasture Mowed	15	11	
16	Oats	15	5	
	Total	695	9671	

The overall computational complexity can be expressed as follows:

$$\begin{aligned}\mathcal{O}(\text{S}^2\text{Mamba}) &= \mathcal{O}(\text{PCS}) + \mathcal{O}(\text{BSS}) \\ &= \mathcal{O}(4N + 2K).\end{aligned} \quad (11)$$

Since the number of pixels and spectral bands in each HSI patch are considerably larger than 4 and 2, respectively, the computational complexity of the proposed S²Mamba is slighter than Transformer-based methods, whose computational complexity is $\mathcal{O}(N^2 + K^2)$, demonstrating its superiority.

IV. DATASETS AND EVALUATION METRICS

In this section, we will present the datasets and evaluation criteria employed in the experiments.

A. Datasets

We conduct evaluations of our S²Mamba on four publicly available datasets, focusing on HSI classification: IP, PU, Houston 2013, and WHU-Hi-LongKou datasets. The experiments on IP, PU, and Houston 2013 use the same training and validate samples as [13], i.e., using a disjoint sampling strategy. The experiments on the WHU-Hi-LongKou dataset use the same training and validated samples as [32] for a fair comparison.

1) Indian Pines: The IP dataset consists of 145 × 145 pixels at a ground sampling distance of 20 m and 220 spectral bands spanning the wavelength range of 400–2500 nm, in which 200 spectral bands are preserved after removing 20 noisy and water absorption bands. The dataset is annotated using 16 land cover categories, including crops, trees, and other vegetation. The details are exhibited in Table I.

2) Pavia University: The PU dataset consists of 610 × 340 pixels at a ground sampling distance of 1.3 m and 103 spectral bands spanning the wavelength range of 430–860 nm, which is annotated using nine land cover categories, including asphalt, meadows, gravel, trees, metal sheets, bare soil, bitumen, bricks, and shadows. The category details are exhibited in Table II.

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TABLE II
LAND-COVER CATEGORIES OF THE PU DATASET AND ITS CORRESPONDING TRAINING AND TESTING SAMPLES

No.	Category	Training	Testing	Color
1	Asphalt	548	6304	
2	Meadows	540	18146	
3	Gravel	392	1815	
4	Trees	524	2912	
5	Metal Sheets	265	1113	
6	Bare Soil	532	4572	
7	Bitumen	375	981	
8	Bricks	514	3364	
9	Shadows	231	795	
	Total	3921	40002	

TABLE III
LAND-COVER CATEGORIES OF THE HOUSTON 2013 DATASET AND ITS CORRESPONDING TRAINING AND TESTING SAMPLES

No.	Category	Training	Testing	Color
1	Healthy Grass	198	1053	
2	Stressed Grass	190	1064	
3	Synthetic Grass	192	505	
4	Tree	188	1056	
5	Soil	186	1056	
6	Water	182	143	
7	Residential	196	1072	
8	Commercial	191	1053	
9	Road	193	1059	
10	Highway	191	1036	
11	Railway	181	1054	
12	Parking Lot1	192	1041	
13	Parking Lot2	184	285	
14	Tennis Court	181	247	
15	Running Track	187	473	
	Total	2832	12197	

3) Houston 2013: The Houston 2013 dataset consists of 349 × 1905 pixels at a ground sampling distance of 2.5 m and 144 spectral bands spanning the wavelength range of 380–1050 nm. The dataset is annotated using 15 land cover categories, including healthy grass, stressed grass, synthetic grass, trees, soil, water, residential, and others. The details are exhibited in Table III.

4) WHU-Hi-LongKou: The WHU-Hi-LongKou dataset consists of 550 × 400 pixels along with 270 spectral bands spanning the wavelength range of 0.4–1.0 μm. The dataset is annotated using nine land cover categories, including corn, cotton, sesame, broad-leaf soybean, narrow-leaf soybean, rice, water, roads and houses, and mixed weed. The details are exhibited in Table IV.

B. Evaluation Metrics

In this article, we utilized three commonly used evaluation metrics: OA, average accuracy (AA), and kappa coefficient (κ) to evaluate the classification performance of our S^{2}Mamba.

V. EXPERIMENT

A. Implementation Details

In this study, all the experiments are conducted within the PyTorch framework, utilizing a single NVIDIA GeForce RTX 4090 with 24-GB GPU memory. For the initialization of our S^{2}Mamba, its parameters are randomly initialized by a zero-mean normal distribution with a standard deviation of 0.01. AdamW [66] is adopted as the optimizer for S^{2}Mamba, where the exponential learning rate strategy is applied with an initial value of 0.0001. The model is trained with 400 epochs, where the batch size is set to 64. The patch size of inputs is set to 7, 11, 9, and 9 for IP, PU, Houston 2013, and WHU-Hi-LongKou, respectively. The number of blocks N is set to 1. The threshold τ is set to 0.1.

TABLE IV
LAND-COVER CATEGORIES OF THE WHU-HI-LONGKOU DATASET AND ITS CORRESPONDING TRAINING AND TESTING SAMPLES

No.	Category	Training	Testing	Color
1	Corn	25	34486	
2	Cotton	25	8349	
3	Sesame	25	3006	
4	Broad-Leaf Soybean	25	63187	
5	Narrow-Leaf Soybean	25	4126	
6	Rice	25	11829	
7	Water	25	67031	
8	Roads and Houses	25	7099	
9	Mixed Weed	25	5204	
	Total	3921	40002	

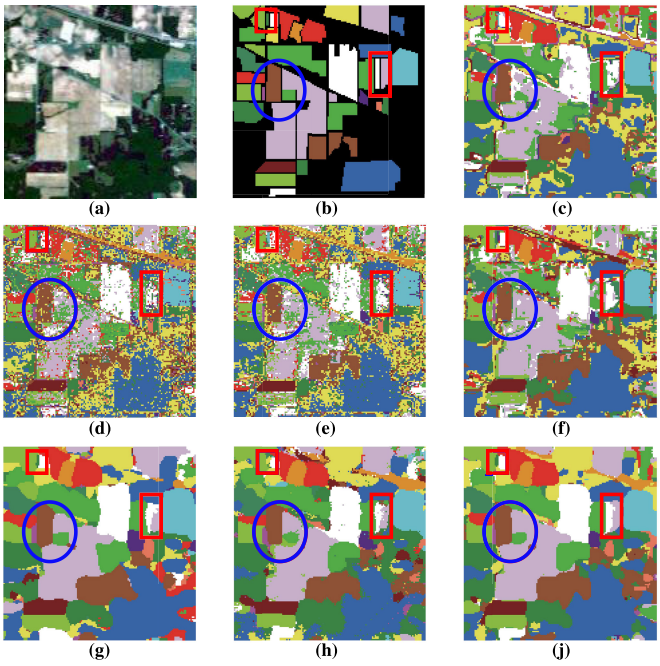

Fig. 3. Qualitative classification results on the IP dataset. (a) False-color Map. (b) Ground truth. (c) miniGCN [12]. (d) CasRNN [31]. (e) ViT [20]. (f) SpectralFormer [13]. (g) SSFTT [14]. (h) GraphGST [32]. (j) S^{2}Mamba.

B. Comparison With State of the Arts

We conduct a comprehensive evaluation of our S^{2}Mamba and compare it with several comparative approaches, including CNN-based (1-D CNN, 2-D CNN, and miniGCN (TGRS'2020) [12]), RNN-based (RNN and CasRNN (TGRS'2019) [31]), and Transformer-based (ViT [20], Spectralformer (TGRS'2021) [13], SSFTT (TGRS'2022) [14], morphformer (TGRS'2023) [15], and GraphGST

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TABLE V
COMPARISON WITH OTHER STATE-OF-THE-ART METHODS IN TERMS OF (OA%/AA%/κ) ON THE IP TESTING SET. THE BEST IS IN BOLD

Category	CNNs			RNN		Transformers					Mamba
	1-D CNN	2-D CNN	miniGCN	RNN	CasRNN	ViT	SpectralFormer	morphFormer	SSFTT	GraphGST	**S²Mamba**
1	47.83	65.90	72.54	69.00	61.78	53.25	70.52	93.14	91.18	**95.81**	94.44
2	42.35	76.66	55.99	58.93	57.78	66.20	81.89	97.70	98.72	98.85	**100.00**
3	60.87	92.39	92.93	77.17	75.00	86.41	91.30	100.00	100.00	100.00	**100.00**
4	89.49	93.96	92.62	82.33	90.16	89.71	95.53	96.87	96.19	97.09	**98.43**
5	92.40	87.23	94.98	67.72	81.35	87.66	85.51	99.86	100.00	98.57	**100.00**
6	97.04	97.27	98.63	89.07	87.70	89.98	99.32	99.77	100.00	99.54	**100.00**
7	59.69	77.23	64.71	69.06	79.08	72.22	81.81	86.71	95.10	97.93	**98.47**
8	65.38	57.03	68.78	63.56	56.29	66.00	75.48	97.93	94.44	94.50	**98.10**
9	93.44	72.87	69.33	65.07	60.11	57.09	73.76	94.33	90.96	95.04	**95.04**
10	99.38	100.00	98.77	95.06	95.06	97.53	98.77	100.00	100.00	98.76	**100.00**
11	84.00	92.85	87.78	88.67	82.23	87.62	93.17	99.12	99.35	**99.59**	97.67
12	86.06	88.18	50.00	50.00	46.97	63.94	78.48	99.09	100.00	98.48	**100.00**
13	91.11	100.00	100.00	97.78	97.78	95.56	100.00	100.00	100.00	100.00	**100.00**
14	84.62	84.62	48.72	66.67	56.41	79.49	79.49	92.31	100.00	100.00	**100.00**
15	100.00	100.00	72.73	81.82	81.82	90.91	100.00	100.00	100.00	100.00	**100.00**
16	80.00	100.00	80.00	100.00	100.00	80.00	100.00	100.00	100.00	100.00	**100.00**
OA (%)	70.43	75.89	75.11	70.66	68.65	71.86	81.76	96.38	96.11	97.06	**97.92**
AA (%)	79.60	86.64	78.03	76.37	75.59	78.97	87.81	97.30	97.92	98.39	**98.88**
κ	0.6642	0.7281	0.7164	0.6673	0.6464	0.6804	0.7919	0.9584	0.9555	0.9664	**0.9761**

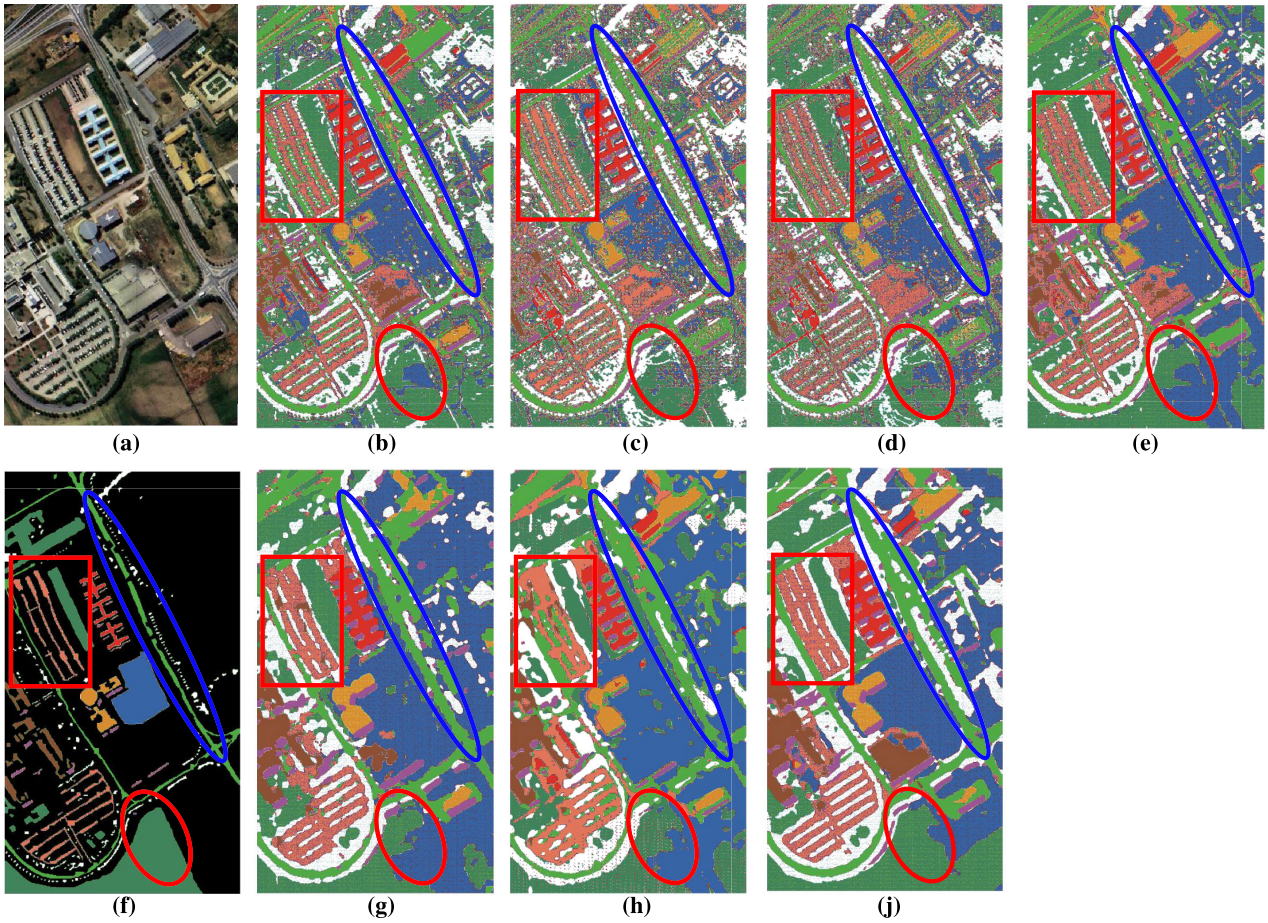

Fig. 4. Qualitative classification results on the PU dataset. (a) False-color Map. (b) miniGCN [12]. (c) CasRNN [31]. (d) ViT [20]. (e) SpectralFormer [13]. (f) Ground truth. (g) SSFTT [14]. (h) GraphGST [32]. (j) S²Mamba.

(TGRS'2024) [32]) methods, where 1-D CNN, 2-D CNN, RNN, and ViT are implemented following [13]. All methods are tested using the optimal experimental settings reported in their papers or reimplemented by their official code.

1) Indian Pines: In Table V, we benchmark performances on the IP dataset. The results, presented in Table V, demonstrate that our approach significantly outperforms state-of-the-art HSI classification methods on 14 categories, achieving the best comprehensive performance in terms of OA (97.92% versus 97.06%), AA (98.88% versus 98.39%), and κ (0.9761 versus 0.9664). In particular, we surpass the performance of the typical Transformer-based method,

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TABLE VI

COMPARISON WITH OTHER STATE-OF-THE-ART METHODS IN TERMS OF (OA%/AA%/κ) ON THE PU TEST SET. THE BEST IS IN BOLD

Category	CNNs			RNN		Transformers					Mamba
	1-D CNN	2-D CNN	miniGCN	RNN	CasRNN	ViT	SpectralFormer	morphFormer	SSFTT	GraphGST	**S^{2}Mamba**
1	88.90	80.98	**96.35**	84.01	77.62	71.51	82.73	89.90	87.64	84.99	96.24
2	58.81	81.70	89.43	66.95	63.41	76.82	94.03	75.26	76.60	82.43	**98.75**
3	73.11	67.99	**87.01**	58.46	57.30	46.39	73.66	85.90	85.56	79.94	85.95
4	82.07	97.36	94.26	97.70	98.42	96.39	93.75	86.33	95.54	90.80	**97.73**
5	99.46	99.64	99.82	99.10	99.37	99.19	99.28	95.87	**100.00**	100.00	99.10
6	97.92	97.59	43.12	83.18	75.17	83.18	90.75	95.54	98.08	90.66	**99.30**
7	88.07	82.47	90.96	83.08	88.48	83.08	87.56	98.27	99.18	98.78	**99.90**
8	88.14	97.62	77.42	89.63	87.25	89.63	95.81	98.01	94.41	93.10	**98.93**
9	99.87	95.60	87.27	96.48	99.11	96.48	94.21	96.86	98.62	**99.25**	98.36
OA (%)	75.50	86.05	79.79	77.13	73.60	76.99	91.07	84.65	85.72	86.39	**97.81**
AA (%)	86.26	88.99	85.07	84.29	82.91	80.22	90.20	91.32	92.85	91.11	**97.14**
κ	0.6948	0.8187	0.7367	0.7101	0.6677	0.7010	0.8805	0.8162	0.7941	0.8220	**0.9705**

TABLE VII

COMPARISON WITH OTHER STATE-OF-THE-ART METHODS IN TERMS OF (OA%/AA%/κ) ON THE HOUSTON 2013 TESTING SET. THE BEST IS IN BOLD

Category	CNNs			RNN		Transformers					Mamba
	1-D CNN	2-D CNN	miniGCN	RNN	CasRNN	ViT	SpectralFormer	morphFormer	SSFTT	GraphGST	**S^{2}Mamba**
1	87.27	85.09	**98.39**	82.34	82.62	82.81	81.86	93.63	85.37	85.66	83.10
2	98.21	99.91	92.11	94.27	96.90	96.62	100.00	98.02	99.81	99.91	**100.00**
3	**100.00**	77.23	99.60	99.60	99.60	99.80	95.25	99.20	95.05	95.64	99.60
4	92.99	97.73	96.78	97.54	96.97	**99.24**	96.12	95.64	90.24	91.29	98.20
5	97.35	99.53	97.73	93.28	97.35	97.73	99.53	98.95	100.00	100.00	**100.00**
6	95.10	92.31	95.10	95.10	95.10	95.10	94.41	100.00	100.00	**100.00**	95.80
7	77.33	**92.16**	57.28	83.77	76.21	76.77	83.12	86.66	75.00	76.96	89.37
8	51.38	79.39	68.09	56.03	44.63	55.65	76.73	83.85	89.74	**89.84**	88.60
9	27.95	86.31	53.92	72.14	64.97	67.42	79.32	75.92	76.86	83.57	**92.45**
10	90.83	43.73	77.41	84.17	78.28	68.05	78.86	66.50	90.92	91.79	**92.57**
11	79.32	87.00	84.91	82.83	88.43	82.35	88.71	82.06	87.95	91.46	**91.56**
12	76.56	66.28	77.23	70.61	66.38	58.50	87.32	87.22	**92.02**	90.49	90.97
13	69.47	**90.18**	50.88	69.12	70.53	60.00	72.63	84.91	80.00	81.40	89.12
14	99.19	90.69	98.38	98.79	100.00	98.79	100.00	100.00	100.00	100.00	**100.00**
15	98.10	77.80	98.52	95.98	96.62	98.73	99.79	99.57	100.00	100.00	**100.00**
OA (%)	80.04	83.72	81.71	83.23	81.22	80.41	88.01	88.27	89.62	90.80	**93.36**
AA (%)	82.74	84.35	83.09	85.04	83.64	82.50	88.91	90.15	90.87	91.93	**94.09**
κ	0.7835	0.8231	0.8018	0.8183	0.7967	0.7876	0.8699	0.8727	0.8873	0.8853	**0.9279**

TABLE VIII

COMPARISON WITH OTHER STATE-OF-THE-ART METHODS IN TERMS OF (OA%/AA%/κ) ON THE WHU-HI-LONGKOU TESTING SET. THE BEST IS IN BOLD

Category	CNNs			RNN		Transformers					Mamba
	1-D CNN	2-D CNN	miniGCN	RNN	CasRNN	ViT	SpectralFormer	morphFormer	SSFTT	GraphGST	**S^{2}Mamba**
1	93.81	99.44	96.29	87.88	81.69	63.85	95.80	96.14	98.49	98.45	**100.00**
2	84.88	73.54	44.71	39.86	47.63	38.42	65.61	93.28	97.82	89.95	**99.32**
3	89.09	91.68	89.16	61.71	63.84	40.19	76.74	98.36	99.27	99.73	**100.00**
4	85.20	92.45	81.61	72.48	66.93	69.45	81.77	85.78	92.28	94.63	**97.87**
5	82.89	78.55	75.50	69.75	67.74	58.07	80.92	91.03	97.02	99.20	**99.59**
6	95.76	92.54	96.13	89.97	90.17	89.33	95.22	99.53	96.17	99.51	**99.85**
7	99.89	99.97	99.78	**99.99**	99.97	99.96	99.59	99.82	98.63	99.97	98.98
8	91.56	82.88	91.48	84.94	83.53	75.17	81.94	92.92	91.73	**96.08**	95.15
9	91.78	87.34	80.75	57.97	59.59	28.46	85.53	90.94	97.79	94.62	**98.29**
OA (%)	92.47	94.58	89.69	83.63	81.18	76.89	90.12	93.91	96.18	97.34	**98.75**
AA (%)	90.54	88.71	83.93	73.84	73.45	62.54	84.79	94.20	96.58	96.91	**98.78**
κ	0.9027	0.9289	0.8671	0.7907	0.7612	0.7071	0.8725	0.9212	0.9503	0.9652	**0.9836**

i.e., SpectralFormer [13], which utilizes transformer architecture to extract long-range dependencies from continuous spectral bands. In contrast, our S^{2}Mamba, involving more efficient basic structures and elaborate designs, achieves superior results in terms of OA, AA, and κ (e.g., improving the OA from 81.76% to 97.92%). Furthermore, Fig. 3 shows examples of prediction maps. It demonstrates that our S^{2}Mamba is capable of producing accurate predictions of each category.

2) Pavia University: We further evaluate our S^{2}Mamba on the PU dataset, whose scenario includes numerous complex spatial textures. As illustrated in Table VI, our method significantly outperforms the comparative techniques with at least 6.74%, 4.29%, and 9.00% in terms of OA, AA, and κ,

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respectively. This confirms that our S²Mamba can offer a more satisfactory solution under complex scenarios. In particular, our S²Mamba outperforms it with 6.74%, 6.94%, and 9.00% in terms of OA, AA, and κ, respectively, compared to the typical Transformer-based method [13]. In comparison with the most recent method, i.e., GraphGST [32], our method achieves an 11.42% gain in terms of OA, which demonstrates the effectiveness of our proposed S²Mamba.

Fig. 4 shows that our approach achieves superior performance than others with more completed prediction maps, such as asphalt (masked in light green) and meadow (masked in dark green).

3) Houston 2013: Furthermore, we compare our proposed method with the state-of-the-art methods on the more challenging dataset, i.e., Houston 2013. The experimental results, presented in Table VII, demonstrate that our S²Mamba significantly outperforms state-of-the-art HSI classification methods on seven categories, achieving the best comprehensive performance in terms of OA, AA, and Kappa coefficient. Specifically, it exhibits 2.56%, 2.16%, and 4.26% gains in comparison with the advanced Transformer-based method [32] on OA, AA, and κ, respectively. Fig. 5 shows classification predictions, which demonstrates that our S²Mamba realizes the most precise results. For instance, our method is capable of accurately identifying highways (masked in blue) under shadows, demonstrating its effectiveness.

4) WHU-Hi-LongKou: Moreover, we make a comparison between our proposed method and other advanced methods on a more recent dataset, i.e., WHU-Hi-LongKou. As illustrated in Table VIII, the experimental results demonstrate that our S²Mamba significantly outperforms state-of-the-art HSI classification methods on nearly all categories, achieving the best comprehensive performance in terms of OA, AA, and Kappa coefficient. Specifically, it exhibits 1.41%, 1.87%, and 1.84% gains in comparison with the state-of-the-art method GraphGST [32] on OA, AA, and κ, respectively. Fig. 6 reveals classification predictions, which further confirms that our S²Mamba realizes the most precise results. For example, our method exhibits more remarkable recognition performance on corn, roads (masked in the red rectangle), and broad-leaf soybeans (masked in the blue rectangle), demonstrating its superiority.

In addition, to verify the efficiency of the proposed S²Mamba, we carried out a comparison of model parameters and running times on the IP, PU, and Houston 2013 datasets. For a fair comparison, all experiments are conducted in the same environment. As shown in Fig. 7, in comparison with the typical Transformer-based HSI classification model [13], our method can greatly reduce the model parameters and seriously raise the training speed. This confirms the efficiency and lightweight of S²Mamba, which is consistent with our motivation.

C. Ablation Study

To demonstrate the effectiveness of each component in S²Mamba, we perform a thorough ablation study on IP, PU, Houston 2013, and WHU-Hi-LongKou datasets. Identical training strategies are employed in all the experiments.

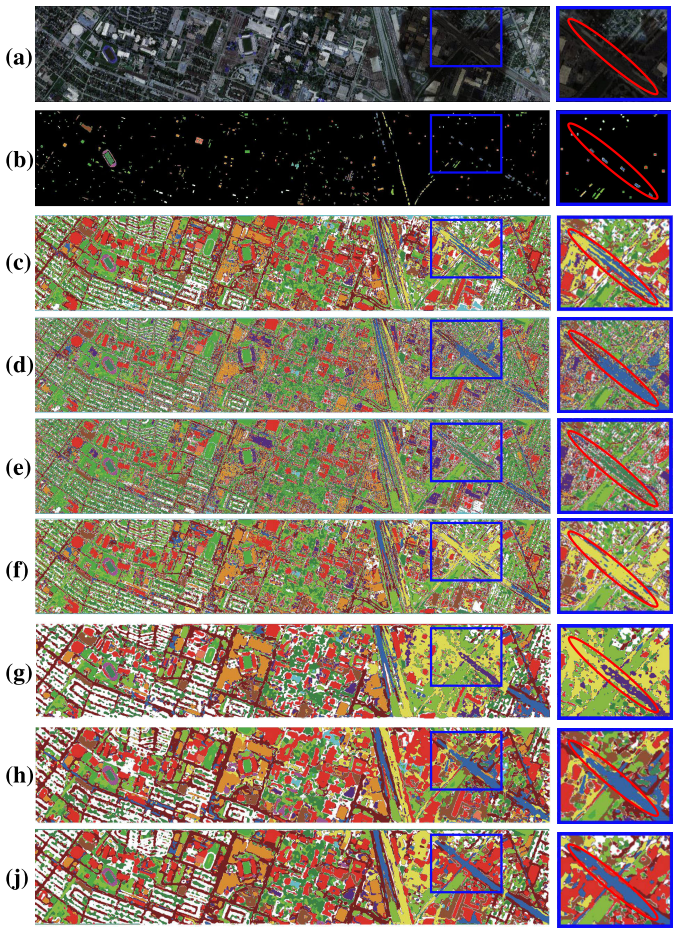

Fig. 5. Qualitative classification results on the Houston 2013 dataset. (a) False-color Map. (b) Ground Truth. (c) miniGCN [12]. (d) CasRNN [31]. (e) ViT [20]. (f) SpectralFormer [13]. (g) SSFTT [14]. (h) GraphGST [32]. (j) S²Mamba.

Table IX shows the impact of each component in S²Mamba. The first row shows the performance using the PCS mechanism, which efficiently considers the spatial relations of pixels by a Mamba-based module. As can be seen, it outperforms most comparative methods with 96.45%, 96.42%, 90.78%, and 97.90% OA on IP, PU, Houston 2013, and WHU-Hi-LongKou datasets, respectively, due to its effective ability in modeling spatial contexts. Then, by incorporating a BSS mechanism, we improve performance to 97.19%, 97.17%, 92.74%, and 98.52% OA on the four datasets. This suggests that the BSS can offer more discriminative cues by scanning continuous spectral bands. Finally, by combining both the PCS, BSS, and SMG, we further boost the performance to 97.92%/98.80%/0.9746, 97.81%/97.14%/0.9705, 93.36%/94.09%/0.9279, and 98.75%/98.78%/0.9836 in terms of OA/AA/κ on the four datasets, which confirms the SMG module can preferably merge spatial and spectral features for each location by a feature competition. The comprehensive results demonstrate that our S²Mamba

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TABLE IX

ABLATION STUDY ON THE IMPACT OF EACH COMPONENT IN S^{2}MAMBA. "PCS": PATCH CROSS SCANNING. "BSS": BIDIRECTIONAL SPECTRAL SCANNING. "SMG": SPATIAL–SPECTRAL MIXTURE GATE

PCS	BSS	SMG	Indian Pines			Pavia University			Houston 2013			WHU-Hi-LongKou		
			OA (%)	AA (%)	κ	OA (%)	AA (%)	κ	OA (%)	AA (%)	κ	OA (%)	AA (%)	κ
✓			96.45	97.99	0.9593	96.42	96.53	0.9521	90.78	92.05	0.9002	97.90	98.36	0.9726
	✓		96.51	98.27	0.9599	96.24	96.21	0.9493	92.19	93.29	0.9154	97.42	97.78	0.9663
✓	✓		97.19	98.56	0.9678	97.17	97.12	0.9620	92.74	93.36	0.9211	98.52	98.58	0.9806
✓	✓	✓	**97.92**	**98.88**	**0.9761**	**97.81**	**97.14**	**0.9705**	**93.36**	**94.09**	**0.9279**	**98.75**	**98.78**	**0.9836**

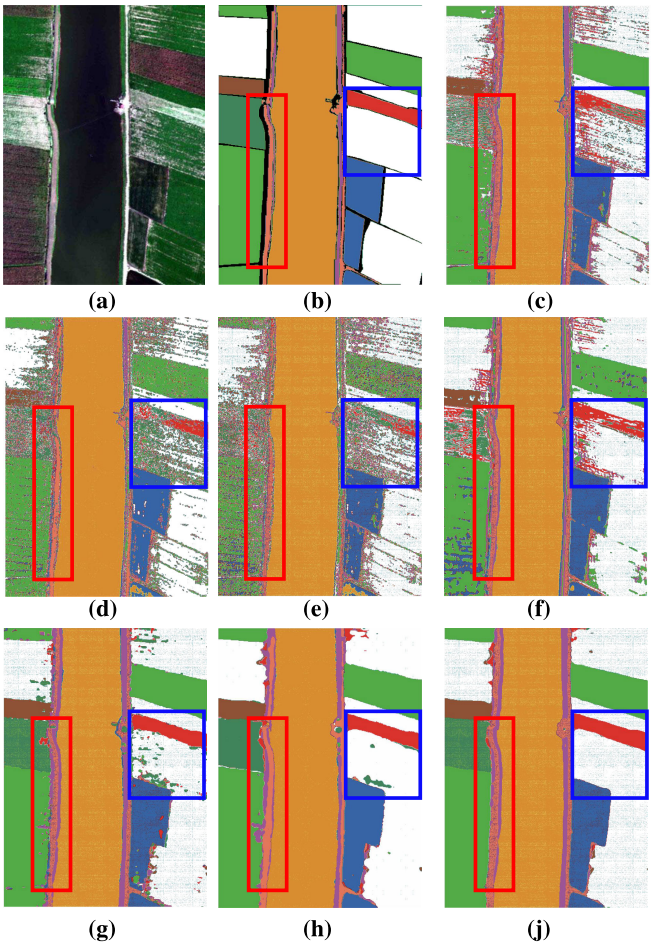

Fig. 6. Qualitative classification results on the WHU-Hi-LongKou dataset. (a) False-color map. (b) Ground truth. (c) miniGCN [12]. (d) CasRNN [31]. (e) ViT [20]. (f) SpectralFormer [13]. (g) SSFTT [14]. (h) GraphGST [32]. (j) S^{2}Mamba.

can effectively learn discriminative representations of HSI data.

To further analyze the effectiveness of each component in S^{2}Mamba, we visualize the feature distribution under different model settings in Fig. 8 using t-SNE [67]. It can be seen that each component of S^{2}Mamba can greatly improve the discriminative ability of the classification model, especially for spatial–spectral feature fusion. When jointly applying BSS and PCS modules, the feature disparity of S^{2}Mamba significantly increases. After inserting all the modules, S^{2}Mamba achieves the best performance, demonstrating the significance of each component.

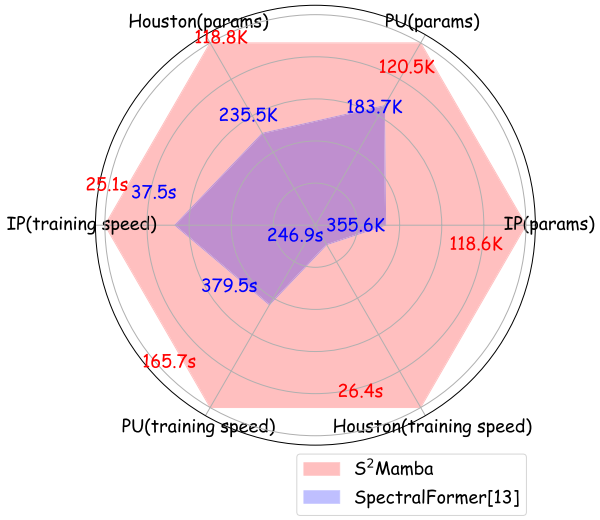

Fig. 7. Comparison of model parameters and running times on the IP, PU, and Houston 2013 (Houston) datasets.

D. Parameter Analysis

In this section, we conduct comprehensive experiments to analyze the hyperparameters within the proposed S^{2}Mamba. The remaining hyperparameters are consistently set to optimal values across all experiments. All the datasets are employed to demonstrate the robustness of the proposed method on fluctuations of hyperparameters.

First, we study the effectiveness of hyperparameters in S^{2}Mamba, including patch size, gating threshold, latent dimension, and layer numbers. As shown in Fig. 9(a), it can be observed that the optimal patch sizes of IP, PU, Houston 2013, and WHU-Hi-LongKou are 7, 11, 9, and 9, separately, which is consistent with the expectation that the last three datasets comprise more complex spatial boundaries, thereby larger patch inputs are required. Fig. 9(b) demonstrates that firming the gating threshold to 0.1 can achieve satisfactory results, which can filter those redundant features. Fig. 9(c) and (d) denotes that 64 and 1 are the optimal values of hidden dimension and layer number, respectively. This confirms our S^{2}Mamba can achieve state-of-the-art performance with a lightweight network, whereas deeper layers or larger hidden dimensions cannot yield additional improvements.

Furthermore, to explore the impact of different scanning mechanisms in S^{2}Mamba, we conduct an experimental comparison of various scanning modes on the IP dataset, including forward spectral scanning (FSS), reverse spectral scanning

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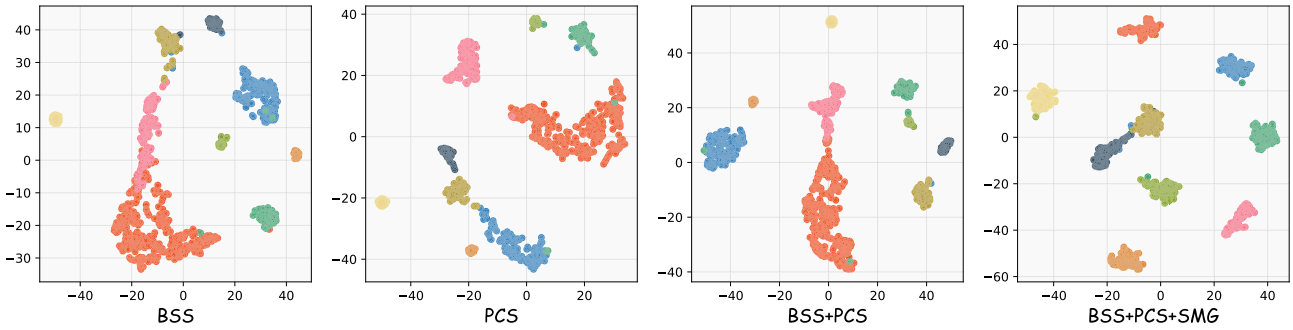
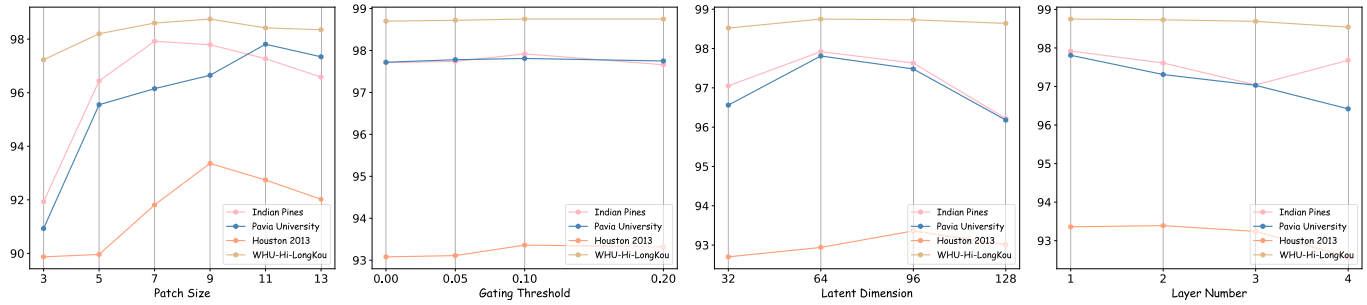


Fig. 8. Feature visualization of different components in S²Mamba using t-SNE [67] on the test set of the PU dataset.



(a) (b) (c) (d)

Fig. 9. Parameter analysis of S²Mamba. (a) Parameter analysis of different patch sizes. (b) Parameter analysis of different gating thresholds. (c) Parameter analysis of different latent dimensions. (d) Parameter analysis of different layer numbers.

TABLE X

IMPACT OF VARIOUS SCANNING MECHANISMS IN S²MAMBA. "FSS": FORWARD SPECTRAL SCANNING. "RSS": REVERSE SPECTRAL SCANNING. "BSS": BIDIRECTIONAL SPECTRAL SCANNING. "SCS": SURROUNDING-TO-CENTER SCANNING. "CSS": CENTER-TO-SURROUNDING SCANNING. "PCS": PATCH CROSS SCANNING

Scanning Mechanism	OA (%)	AA (%)	κ
FSS	95.76	97.82	0.9514
RSS	96.18	98.06	0.9563
BSS	96.51	98.27	0.9599
BSS+SCS	96.94	98.47	0.9649
BSS+CSS	96.95	98.48	0.9650
BSS+PCS	97.92	98.88	0.9761

(RSS), BSS, surrounding-to-center scanning (SCS), center-to-surrounding scanning (CSS), and PCS, whose scanning routes are as shown in Fig. 10. As illustrated in Table X, it can be seen that the BSS mechanism outperforms other spectral scanning strategies with at least 0.33%, 0.21%, and 0.36% gains in terms of OA, AA, and κ, respectively. Moreover, the PCS mechanism exhibits more satisfactory performance compared to surrounding-to-center and center-to-surrounding scanning mechanisms in terms of spatial scanning. Overall, our employed mechanism (BSS + PCS) can greatly outperform others due to its full consideration of global spatial–spectral features, which demonstrates its effectiveness.

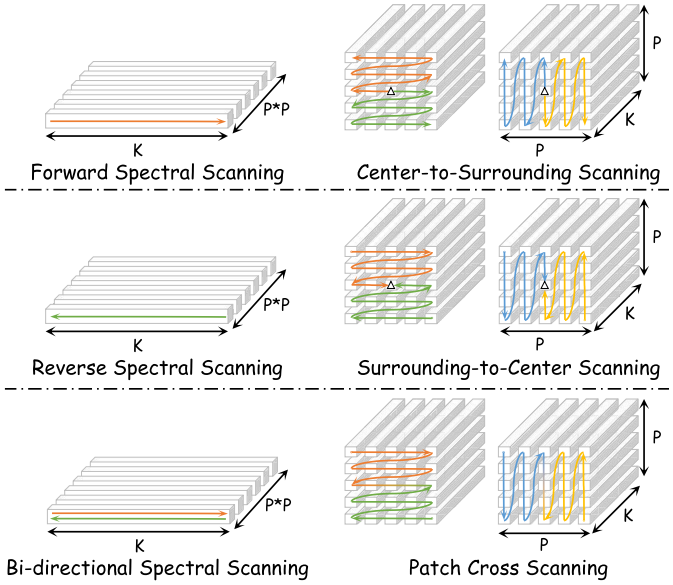


Fig. 10. Visualization of different scanning mechanisms in S²Mamba.

VI. CONCLUSION

In this article, we proposed S²Mamba, a novel architecture for HSI classification. S²Mamba comprises a PCS mechanism and a BSS mechanism for learning the contextual information from spatial and spectral aspects, respectively, which utilize the selective structured SSMs as alternatives to self-attention mechanisms for capturing long-range dependency with linear complexity, thereby efficiently improving the results.

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Furthermore, to optimally merge the above features, an SMG was proposed to adjust the fusion ratio for each position with learnable matrices, further enhancing the classification performance. Finally, we evaluated the proposed S^{2}Mamba on four public HSI classification datasets, and the experimental results verified its superiority.

Throughout this article, we investigated a novel HSI classification architecture based on SSMs, while more potential tasks in HSI analysis remain uncovered. Considering the efficiency and effectiveness of the Mamba models, we look forward to generalizing S^{2}Mamba to improve other tasks in HSI in the future.

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